NAME : .....

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# RACE

Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2024
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | MATHEMATICS | M.M. 64
REVISION ASSINGMENT # 17 | (DE + AUC) | TIME : 60 MIN.

## SECTION–I

**Straight Objective Type** (3 Marks each, –1 for wrong answer)

**1.** Area bounded by $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ and $y = 3x - 15$, for $x > 1$, is -

(A) 73 (B) 13/6 (C) 73/6 (D) none of these

**2.** A tangent line with positive slope to a curve $y = x^2 + 1$ is drawn such that area bounded by curve, tangent line and co-ordinate axes is $\frac{11}{3}$ then x co-ordinate of point of tangency is

(A) 5 (B) 3 (C) 7 (D) 1

**3.** Consider a function $f(x) = \int_0^x [\cos t]\,dt$, $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ then area bounded by curve $y = f(x)$, x axis and lines $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ is (where [.] denotes greatest integer function) :-

(A) $\pi$ (B) $\pi^2$ (C) $\frac{\pi^2}{2} - 1$ (D) $\frac{\pi^2}{2}$

**4.** The area of the smaller portion above x-axis bounded by the curves $y^2 = 8x$ and $\frac{(x-2)^2}{4} + \frac{y^2}{16} = 1$, is -

(A) $4\pi - \frac{32}{3}$ (B) $8\pi - \frac{16}{3}$ (C) $8\pi - \frac{64}{3}$ (D) $2\pi - \frac{16}{3}$

**5.** The general solution of the differential equation $\frac{dy}{dx} = \left(x^3 - 2x\tan^{-1} y\right)\left(1 + y^2\right)$ is -

(A) $2\tan^{-1}x = y^2 - 1 + 2ce^{-x^2}$ (B) $2\tan^{-1}y = x^2 - 1 + 2ce^{-x^2}$

(C) $2\tan^{-1}y = y^2 - 1 + 2ce^{-x^2}$ (D) $2\tan^{-1}x = x^2 - 1 + 2ce^{-y^2}$

**6.** If $y = e^{(K+1)x}$ is a solution of differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$, then k =

(A) –1 (B) 0 (C) 1 (D) 2

**7.** The order and degree of the differential equation $\sqrt[3]{\frac{dy}{dx}} - 4\frac{d^2y}{dx^2} - 7x = 0$ are a and b, then a + b is -

(A) 3 (B) 4 (C) 5 (D) 6

**8.** The area bounded by the x-axis, part of the curve $y = 1 + x^{-2}$ and the ordinates $x = 1$ & $x = 2$ is divided into equal parts by the ordinate at $x = a$ such that $a = \frac{3 + \sqrt{p}}{8}$, then the value of p is -

(A) 70 (B) 71 (C) 72 (D) 73

**Multiple Correct Answer Type** (4 Marks each, –1 for wrong answer)

**9.** The general solution of the differential equation $e^x dx = \sqrt{1 - e^{2x}}\,dy$ is -

(A) $y = K - \cos^{-1}(e^x)$ (B) $y = \sin^{-1}(e^x) + C$

(C) $y = \sin^{-1}\sqrt{1 - e^{2x}} + C$ (D) $y = \cos^{-1}\sqrt{1 - e^{2x}} + C$



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**10.** Solution of the differential equation $b(x^2 + a^2)dy + a(y^2 + b^2)dx = 0$ is -

(A) $bx + ay = c(ab - xy)$ (B) $\tan^{-1}\left(\frac{x}{a}\right) + \tan^{-1}\left(\frac{y}{b}\right) = c$

(C) $\cot^{-1}\left(\frac{a}{x}\right) - \tan^{-1}\left(\frac{y}{b}\right) = c$ (D) $\tan^{-1}\left(\frac{x}{a}\right) - \cot^{-1}\left(\frac{b}{y}\right) = c$

**11.** A function $y = f(x)$ satisfies the differential equation $\frac{d}{dx}(xy) = 1$ and $f(1) = 2$, then :-

(A) $f$ periodic function (B) $f$ is one-one function

(C) domain is of $f$ is $R - \{0\}$ (D) function is neither odd nor even

**12.** If A denotes the area between the curve $2\{y\} = [x] + 1$ for $0 \le y < 1$ and the x-axis between $x^2 - x \le 0$, then A is less than or equal to (where $\{.\}$ and $[.]$ are the fractional part and greatest integer functions respectively) -

(A) $\frac{\pi}{6}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

**13.** Area enclosed by the curve $y = \sin x$ with x-axis between $x = 2n\pi$ to $x = 2(n+1)\pi$, is-

(A) $\int_0^{2\pi} \sin x\,dx$ (B) $2\int_0^{\pi} \sin x\,dx$ (C) $4\int_0^{\pi/2} \sin x\,dx$ (D) 4

## SECTION–II

**Numerical Grid Type (Upto Second Decimal place)** (4 Marks each, –1 for wrong answer)

**1.** If the area bounded by x-axis, y-axis, $x = 2$, $y=2$ & $xy \le 2 \le 8\,(x^2 + y^2)$ is $\alpha - \frac{\pi}{\beta} + \ell n\gamma$ where $\alpha,\beta,\gamma \in N$ then $\beta - 2(\alpha + \gamma)$ is equal to

**2.** Area bounded by the straight lines $x^2y - y^3 - x^2 + 5y^2 - 8y + 4 = 0$ is equal to (in sq. units)

**3.** If $\frac{dy}{dx} = \frac{e^x(\sin x + \cos x)}{(\tan y + y\sec^2 y)}$; and the curve $y = f(x)$ passes through origin & $(x_1, \pi)$, where $6 < x_1 < 7$, then $\frac{x_1}{\pi}$ is equal to

**4.** If $y = ax^p + bx^{-q}$ where $p,q \in N$ & a,b are constants such that $x^2y'' + 2xy' = 12y$ then 'p+q' is

**5.** If $(2xy - y^2 - y)dx = (2xy + x - x^2)dy$ and $y(1) = 1$, then the value of $12|y(-1)|$ is

**REVISION ASSINGMENT # 16 APPLICATION OF DERIVATIVES-02 MATHEMATICS**

| | Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SECTION-I | A. | A | B | B | A | A | C | B | B | D | C |
| | Q. | 11 | 12 | 13 | 14 | 15 | | | | | |
| | A. | A,B,C,D | B,C | A,D | A,D | D | | | | | |
| SECTION-II | Q. | 1 | 2 | 3 | 4 | 5 | | | | | |
| | A. | 0.50 | 3.00 | 0.73 | 0.25 | 0.50 | | | | | |



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